

Purposefully Nonstandard Cybernetics an alternative to purpose driven control

1:03:18

okay the next talk is going to be by
Bradley Alisia and colleagues
purposefully non-standard cybernetics an
alternative to purpose-driven control
this is going to be a 1our pre-recorded
talk so thank you

enjoy hello my name is Dr Bradley Al and
on behalf of Jesse parent Morgan huff
and Anda Nelson I'm we're from the
orthogon research and education
laboratory and the cybernetics interest
group and I'm going to talk to you today
about per purposefully non-standard
cybernetics an alternative to
purpose-driven control so we start with
a question what is the definition of
behavior and this is varied throughout
history according to the behaviorist
Behavior was identified as a reflex
elicited by stimulus associations the
consequence of reinforcement and the
current motivational state if we go
forward to the cognitive ists which is
more or less the modern view they would
say that behavior is a set of internal
mental states that are responsible for
observable behaviors with gold directed
end States now it's of note that both
behaviorist and cognitivist Views
involve feedback this leads to learning
a notion of what constitutes a state and
predictive capacity so we start with
this paper Rosen blue th all in 1943
Behavior purpose and teeology published
in the philosophy of science and it's a

formative paper in the field of cybernetics so in that paper they have this typology of Behavioral classifications so we're going to return to this typology throughout the talk and we're going to think about how we can reorganize this typology but it basically defines Behavior as a set of nested things the red text Non classes or they're basically all other things other than the uh black thing that we'll be talking about at each rung of this typology so behavior is active it's purposeful it involves feedback which implies some sort of chology or end goal it's also predictive which involves extrapolation and involves an order of prediction so you might have first order second order or third order prediction so to understand what teeology is and where it comes from we start at the Middle Ages where teeology was the end State or the end goal of some behavioral process was thought of as a Divine causation we then move forward to teeology As Natural cause a and then it branches off into two directions and the first was the vitalism of bergson and evolution by natural selection made famous by Darwin leading from Darwin's work we had arar who talked about tonomy which is purpose versus history and we'll talk a little bit about that later in the talk kind of discontinuous to this are two different streams so the first at the top is where behavior is viewed as habit that's by William James and Thorndike that leads us to the an

molecular ontology of Francis Crick and
feedback is complexity which is
summarized by control theory and
dynamical systems at the bottom is our
target paper that we just talked about
these are the views of Rosenbluth
Wiener and this is feedback relative to a
goal this led us to cognitive science
and reinforcement learning at least the
modern versions of this where people
talk about goal directed behavior and
variations on that in molecular biology
such as molecular
vitalism and the homeostatic methodology
of Lebanian and Ginsburg so we'll
talk about a couple papers here and
frame some of this in context so if we
think about Behavior it's it's
essentially a form of functionalism in
the first paper here Rody and Germain
they argue that functionalism is a
pre-scientific relic so they actually
argue that the whole idea of
functionalism is something that comes
from this era of divine causality
biology is not designed so why should we
expect a function and this is something
that of course maybe is assumed about
function that it comes from some sort of
design or comes from some sort of divine
intervention on the other hand toonomic
systems are defined by spatial and
temporal information and constant
iterations Over States so we do have
these toomic systems but include modern
components only select behaviors can be
defined as selection of a goal or
derived from signals from the goal so

that means that we can't necessarily paint Behavior with a broad brush of teleology only some behaviors may be teleological and others maybe not not all behaviors and this includes biological and machine behaviors are goal selecting and while goal directed behaviors seem to be purposeful and it may be true that some of these behaviors that are known as goal directed are purposeful are there Developmental and evolutionary antecedence so are those Place those uh ancestral behaviors or those basal behaviors or behaviors that occur earlier in development are those also purposeful and the answer may be no so Rosen Lu at all Define behavior as any change in an agent with respect to its surroundings this is much broader than the Neuroscience psychological definition and it sets up a contrast between all behaviors and a subset of purposeful behaviors so if a behavior is put under some sort of selection or if a behavior is selection of something in the environment this might lead us to a subset of voluntary behaviors which then some of those may be purposeful and so purposeful is actually purposeful behaviors is actually a subset of all behaviors to demonstrate this we see that clocks both mechanical and biological are not purposeful as there is no final aspirational state mechanical clocks can keep time and you can set an alarm on them but that alarm is imposed by the outside and it's not a

property of that behavior and of itself
further more biological clock runs VI a
physiological processes like a circadian
rhythm and it can be entrained by
external white but that again is not a
final aspirational

State more generally teleology requires a
signal from a goal so it's some sort of
signal from a goal that's specific to
that behavior and purposes progress to
that goal so again with our uh example
of circadian rhythms it's entrained by
light but that light itself is not a
goal it's just something that's out in
the environment and the system entrains
itself to

it we can also see this with brains in
that brains produce behavioral outputs
but the exact mechanism is hard to
pinpoint and this is the Rost and Lal
article that we saw earlier and this
asks the question signal from what to
goal so there's a signal coming from
something to a goal but it's not clear
what that is this is something one of
our co-authors Jesse parent works on
this is a frontier map for teleology and
related complexity Theory Concepts so
the data from this is drawn from the
complex world book by crck hour this is
published in
2024 and it kind of goes over three
different concepts that are interrelated
so we have teleology organized
complexity and emergence and so you can
see that our paper Rosen blue 1943 is at
the top and this is sort of the first
paper that talks about

theology in sort of a scientific and in
a a manner of
complexity and so this is the first
paper that talks about teeology is a
property of complexity and of complex
systems so a lot of papers were
influenced by Rosen Bluth this uh
tileology was inherited by Weiner's book
cybernetics and control in 194
two other papers adaptive systems in uh
by John Holland in 1962 and
organizations and complexity also in
1962 the good regulator of course in
1970 inherited this concept but it also
inherited another concept from
organizations and complexity which is
organized complexity the good regulator
then passes down this concept of
tileology to the book order and chaos in
1993 and there's a direct link between
Rosen Bluth 1943 and the tame concept of
Mike 11 in 2022 now we going back to the
concept of organized complexity this
comes from organizations and complexity
the good regulator in 1970 and the autop
poesis paper in 1974 both share this
intellectual lineage the autop polies
paper lends that organized complexity
concept to this paper on deiv in 1982
get to this book order and chaos in
1993 and then to the tame Concept in
2020 2 finally we have emergence which
starts with the paper by Anderson war is
different in 1972 inherited by the autop
poli's paper 1974 in turn inherited by
the dissipativity paper 1982 in turn
inherited by Order and Chaos 1993 so you
can see that we have not just theology

as a concept in which we've kind of talked about already but also a number of other related Concepts this is 11 uh this is the tame paper and then Janca Ginsburg this is the U paper and so Lan proposes the tame or technological approach to mind everyone Lev's principle here is this cognition all the way down we can see cognition in terms of Minds we can see cognition in terms of molecular phenomena we can see cognition everywhere in between and so behaviors all those behaviors are cognitive to some extent and as a result we get these diverse Minds that have these different goal directed competencies that allow those systems to achieve embodied agents so in Levan's approach you know kology is a very important concept not just for cognitive type behaviors but behaviors that that go all the way down to the molecular scale and maybe even down to the quantum scale second theology is a means to explain why agents develop towards goals and so you know we can look at uh agents such as artificial agents simple organisms like bacteria then we can see that like teeology is a way to explain learn how to move towards goals the next paper is this U paper by Janca at all and in that paper they propose a concept called unlimited associative learning and this concept is a minimal marker for Consciousness so in this paper they also Champion teeology is an intrinsic reinforcement system you have this internally regulated model in service of

attaining external goal andology plays a role in attaining goals so again in both cases there's a strong tendency to talk about goals the strong tendency towards to but we might ask the question is behavior at these different scales and any sort of behavior that sort of motivated towards some sort of function ta

logical and purposeful or is it simply a product of the history of the system so you know we could talk about teleology and purpose and we've kind of set this up that you know this is something that has kind of come out of the natural study of behavior but sometimes you know we don't really fully you know sometimes we ascribe teleology and purpose to things that maybe are just simply a product of the history of a system first example I want to give is the basov jabotinsky reaction or the bz reaction and at the lower left you can see a image of the bz reaction and this is a deterministic internal model that we can model with differential equations and we can solve those differential equations and get interesting solutions that produces a quite chaotic behavioral output or the Dynamics of the system which are sort of the unfolding history of the system so if we take an example of some sort of chemical soup we can achieve both Jabotinsky reactions or bz reactions by setting up this sort of set of differential equations these ordinary differential equations setting up an initial condition and we get these

spiral waves and scroll waves that result and so if we have slight variations in initial condition we end up with very different systems very different types of patterns in these um in these systems and we can describe this with ODEs and one might say okay there's a purpose to this but the purpose isn't necessarily clear when you run this simulation again and again with different types of initial

conditions there's a chaotic behavior at a variational dynamic dynamical output so this is an example of the tip trajectories of BZ waves this is from a paper I wrote in 1993 and it shows how there's a lot of

variation in the pattern but there's a sort of um you know constant aspect of the pattern in terms of the tip trajectories so one might say okay the purpose is to produce these tip trajectories instead of the actual pattern and the interactions which may be true but then of course you know that's something something that I want you to think about as we go through the rest of this

talk we could also plug the Belousov-Zhabotinsky reaction into a cybernetic model with higher level feedbacks and so I talk about this in this dimensions of morphogenesis lecture on the dorm YouTube channel and what I'm getting at in that talk is that there different dimensions of morphogenesis there's one-dimensional morphogenesis two-dimensional morphogenesis

higher dimensional
morphogenesis and so you see this kind
of output and so I'm asking the question
here is this purposeful or is this a
product of sort of a chemical a
dynamical chemical system with a history
and so we can also look at this as a
cybernetic model where we have feedbacks
and we have other sorts of
complexity and you know we can fit it
also into that behavioral uh typology of
roseen blue

okay another

uh another example is this thing called
an allostasis machine so an Al estasis
machine is basically the output of an
internal model and so an internal model
of course as we talked about is
something that has some sort of capacity
for memory and learning and maybe some
sort of output that looks Behavioral or
cognitive in some way so what we can do
is we can take this internal model we
can have an oscillator which is a
stochastic process or a deterministic
process depending on our oscillator and
we can have a memory with
capacity so we have this oscillator that
produces an output and we have sensory
information that can drive the
oscillator

forward this output creates a trajectory
which is an interaction with between
noisy perturbations in a historical
trajectory that's shown in this red
line and the historical trajectory is
stored in the internal model was a
memory with a capacity so as this red

line unfolds there's a memory of it in the internal model and it responds to external perturbations which drive this uh Red Line in different directions and the oscillator tries to recover to the original state and so you get these dynamics that may look purposeful but really have this stochastic and adaptive component we get more into this aspect of dynamical control systems I look at intrinsic motivation in dynamical control systems so this is this tiomkin paper PRX life and so in this paper they talk about how you might model maybe something like an alisasas machine as a set of actions generated without a specific reward signal so in this paper they talk about that as intrinsic motivation they model this with except States and AST actions so an action maps to a number of states and this results in a trajectory forced by observation stochasticity this is entropy so you produce the trajectory there's an observation that you have and that's the entropy of the system and so you maximize actions that increase susceptibility rather than entropy these are noisy observations that we do in perfect observational States and you can compare par arbitrary length sequences of such sequences in the future this gives your internal model the ability to sort of integrate information and to find sort of solutions to problems the idea is that you have the system that can work towards a goal but it isn't

driven by any sort of teleology it's just
driven by this memory component and the
stochastic component and that's all you
need a number of other papers here that
I'm going to talk about I'm going to put
this in the framework of anticipatory
systems from our paper on tony and any
Tristram so to get to this we need to re
kind of reformulate our behavioral
typology so remember we talked about
behavior is having these different
components we have these non- classes so
in this case we're going from Behavior
to active Behavior a purposeful Behavior
but that's contrasted with historical or
markovian behavior then we have feedback
prediction and Order of prediction and
so ERM refined the Rosen blue that all
theological category to Parts purposeful
agentic and historical markovian so if we
go back to the Benny and Friston paper
they talk about the free energy
principle is managing different forms of
gold
directedness and how you need negative
feedback to attain homeostasis there's
purpose in in cases where the goal is
known which is a markovian system and
surprisal was minimized which resolves
uncertainty then we can expect purpose
purpose isn't really just some sort of
magical thing or it's not some sort of
thing that requires is a strict goal
it's just certain conditions under which
our behavioral trajectory exists we can
also look at anticipatory systems it's
the lousy paper and this anticipatory
systems come from a generative model

with depth so this is where we have temporal aspects of the system that lead us to teleology so it's more that the teleology is the history of the system and this depth comes from an internal model so basically if we go back to the internal model we have some process that generates Behavior we have this memory the memory is basically something that stores everything that the system has experienced and that's how we get to a kind of understanding this a little bit more we want to distinguish between a Markovian system and a non-Markovian system so our Markovian system is where you have inter arrival times so there's a series of things coming in in time this is exponentially distributed in this diagram we have sensory information coming in we have some sort of encoding this encoding is a binary Channel with noise where we have these lines that represent some sort of sort of sensory input or perturbation of the system and the inter arrival times the area between these lines so the inter arrival time is obviously not equidistant in this example but it's in between the red and blue lines or the red and red lines and so we can take those in our arrival times and if they're exponentially distributed we have a Markovian system if they're not then we have a non-Markovian system and so we've kind of developed this uh distinction between Markovian and non-Markovian sensory information and how that relates to an information measure in our Gibsonian information

work so gibsonian information is a markovian but can also be non-markovian information processing system and the citation is here so going back to that behavioral typology we have active purposeful feedback prediction and Order of prediction feedback is everything that is circular non-transitive this is the present p past and future your behavior is controlled by a margin of error Corrections of agent location the time t_e relative to a goal position or state and so if we think about this in terms of a feed forward process feed forward process is where the past is your input present is your process and the future is your output in feedback however it's a little bit different your input comes in the past and then you have this present State here and then you have feedback which is the past and that Loops back to your input which means that your input is now the future and so the future is actually present with past feedback so you see the loop here you have input to the present you have this feedback which is the past and it goes to the Future so now you have this future State before the present State and if that keeps going out you get this all output with alternative Futures so your feedback loop can be positive or negative and in this case it doesn't matter what we're interested in is sort of the order of past present and future so we end up with this input we go to the present we go to the past we go to the future we go

back to the present so we're sort of interdigitating different types of causality here instead of just having this past present and future aspect to things we have this sort of convoluted aspect of past present and future if we go back to our historical map we recall that non olical things can be characterized as feedback as complexity and so there are two papers here that talk about pit control that kind of get us at that idea so pit control is proportional integral derivative control and if we think back to our example of chemotaxis and phototaxis we find that PID control can be applied to chemotaxis and other types of biological control so this is The Balter and Buckley paper they talk about this in terms of a variational free energy minimization given a linear generative model they talk about pit control resembling a reflex arcate mechanism so this is the reflex arc we find in sensory motor control and the free energy principle provides a more direct account of control this is where we don't have any hidden states in the process of proprioception so they're focusing on proprioception and but they also talk about chemotaxis and so they kind of give us this sort of uh variational free energy interpretation of this um in go all which is a more engineering take on this the pi and PD

components or the proportional integral
and the proportional derivative
components and directed Network
synchronization are all sort of uh
components of this control
mechanism in particular the pi and PD
components of PID Define present past
and future components in an internal
model control
system we can then also start to think
of feedback as complexity in terms of
nonlinearity and complexity as well so
this paper by Westerberg at all stimulus
history not expectation Drive sensory
prediction errors in a melean cortex
this is from the bio archive so the
author has framed this in terms of
predictive coding so predictive coding
explains cortical responses in two ways
a feed forward model where an internal
model of expected events occurs or in
terms of an internal model of expected
events or a feedforward approach and a
calculation of prediction error for
unexpected stimuli or feedback approach
the latter are actually Global oddballs
which defy expectations in a local
context so in Mouse and macak which are
their two model organisms they see two
types of signal they see a normal
response where the internal model is
matched by little stimuli and a global
Oddball response which is emerged which
emerges as a higher order response to
this normal response we also can look at
this nonlinearity in terms of Behavioral
output in general and then burstiness or
nonnormality of so this paper burstiness

of human physical activities and their characterization and quantifying postural sway Dynamics using burstiness and interevent time distributions kind of gets at both this aspect of Marian systems and a behavioral system that exhibits sort of nonlinear complexity both papers focus on the burstiness of postural sway in postural sway we have this Paradigm where someone stands on a plate and that plate is a force plate that measures the center mass of the individual and then we can measure this and we get this nice dynamical trajectory that has different features that we can analyze so it's a behavioral output that exhibits a lot of nonlinear characteristics and dynamical characteristics

teuchi and Sano focuses on burstiness and postural sway in terms of adult housework and children's play behaviors they studied physical activities of children between ages two and five and they found that burstiness is not an acquired feature it's something that's innate burstiness is present in healthy healthy individuals not in people with movement disorders or impairments they also looked at inter event time distributions and Associated fluctuations so we talked about inner event times as being key to more Marian and non- Marian systems they found that they're bursty so they're these stochastic signals that have a noise component and sometimes this is in function of someone trying to maintain

their balance and sometimes it's just noise that's generated by the nervous system that's added to the stochastic signal in both cases This Is Not Gold directed it's generated by physiological system it's a very complex behavioral signal and it's not explainable by either a pan or gaussian model which means that it has this very complex set of features so if we do some experimental cybernetics on this we can find that if we have the right internal model we can produce very interesting output signals or behavioral signals some of these T kind of look like they could be chological but they're not this is just simply running some stochastic sign one example is just a pure stochastic signal the other one is having a stochastic signal with feedback that then takes that signal and integrates it at a later date so what we have here are two Fe forward processing one is a spway stochastic signal in Blue on the left and then the other is a sum signal that is STO a stochastic delay of that signal or the signal on the right so on the right we have the blue signal you can kind of see it under the red signal so the red signal is the one that's bursty and it's is basically where at random intervals we sum previous output and we add it to the present output so we end up with these bursts and so the delay which is this uh sum signal that we uh that represent our bursts is the buildup of the input signal it produces these bursts or what

we might call avalanches and so we can see that we can produce these nonlinear signals from a simple internal model so this is the classic coopel paper from 1982 it actually has a couple of interesting things to say of cybernetic oriented things so we have this feed forward which is where you have a process that goes from A to B to C to D and we also have these systems that have strong back coupling so we have a process that goes from A to B to C to D then we have these back couplings from D to C C to b b to a and so what's more likely to be the case of like you know actual sort of complex behavior is not the feed forward signal but the strong back coupling this is where we have ubiquitous feedback Str back coupling can be intractable to analysis but produces the most interesting aspect of network interactions and strong back coupling produces things like Collective Behavior emergent properties stochastic outputs and nonlinear outputs or Avalanches which we saw in our experimental cybernetics a few slides back we can think about feedback and we talked about this relationship between past present and future a few slides back and we also talked about maybe some of these types of behaviors that are non-random but maybe structured in a certain way and we can think about feedback just sort to revisit the idea of feedback that roseen blue F all gave us which is that feedback was purposeful so we can ask the question is feedback

purposeful or simply non- random and so we can think of past present and future relationships as a trip around the feedback Bove namely that we can take our input we can take the present and have a feedback loop that gives us past information into a new present when we get a new time point and that ends up giving us a future that can then be recycled so this has implications for prediction we can go past present future to prediction or we can have inputs that produce these sort of differently ordered past present Futures and use that as the output so feedback mechanisms reorder processes systems with a lot of feedback aren't teleology per se maybe they produce outputs that look tele but they're not teleological and so going back to the different orders of prediction which were kind of like the final output of our Behavioral typology what are the different orders of prediction well we have our first second third and fourth or higher order feedback so we can have these feedbacks that have PR past present future but we can Nest them together so that we can have a past of a past of a past contributing to the present or to the Future and so that's really interesting because what it's suggesting is these higher level feedbacks maybe AR also not tele logical but there's some sort of memory or maybe there's some sort of um you know interval and the way it's structured so there's some

structural aspect of feedback that might be really interesting that isn't chological per se but it given some of these other examples might have some structural significance so if we build a a cybernetic model that is really kind of complex and has a lot of interlocked feedbacks and things like that we can get some really interesting results so our different types of higher order feedback you know we can kind of think of these in terms of the derivatives of position so the derivatives of position revisiting that from undergraduate physics you have velocity acceleration jerk snap crackle pop and these are just where you start with velocity and you get a a derivative of velocity which is acceleration you get a derivative of acceleration which is jerk and so on and these can be thought of in this way of having these higher order feedbacks or these reordering is a past present future to give you these really complex Dynamics in cases of derivatives of position and feedback feedback if it goes farther into the past the effects persist farther into the future so if we go back a couple of feedback loops nested on top of one another and we introduce that to the new present or the new future those effects persist further into the future so we get these really long-term effects in a dynamical system instead of just being local how does this translate into the neuroc correlates of behavior well we go

back to our idea about Behavior sort of being natural really logical and we have to think about like kind of how we kind of assume that behavior does magic things so if we get a behavior like this with higher order feedbacks it tends to resemble the sort of magical aspect it could be something like emergence it could be something like burstiness or something like a set of avalanches that occur and so these are things that you know you know give us really interesting Dynamics but are no more sort of magical orological than any other kind of behavior and so more fully connect systems are perspective or might be sentient or conscious that's sort of the idea of does magic things so you know we don't want to us Mis attribute sentience or Consciousness to something that isn't and of course this is something that we've have a problem with with modern AI systems that people think they're sentient conscious when in fact they just have this higher order feedback and this sort of um you know behavior that isn't necessarily what people think it is magic things or you know what we talk about like complexity results from embodied function a good example of this is kinematic chains which we'll talk about in a little bit but now we introduced this concept of convolutional Cy cybernetic architectures or ccas and so we go back to our behavioral typology and we ask the question can we better understand the role of purpose or meaning and

feedback or temporal order so here we have active Behavior purposeful Behavior theological Behavior and the non-classes of those are nonpurposeful the random behavior and non-feedback or non-phological Behavior respectively but we're going to reorder this behavioral classification where now we have non-prediction leading to the order of complexity rather than prediction leading to the order of prediction so we start off with behavior that's active instead of passive then we have nonpurposeful or random Behavior instead of purposeful and non nonactive or passive and purpose for know non-classes then we have feedback which is non-teleological versus feedback which is theological theological feedback is now the non-class non-teleological feedback is now the path to the order of complexity and then instead of prediction being extrapolation we'll have non- prediction which is also extrapolation and then we have our order of complexity first second and third order so the more feedback we have the more complexity we have so this is an example of a convolutional cybernetic architecture we basically have this toy example where we have an input and an output the input is 0000 0 the output is 111 and we have these boxes which process the present to the Future and then our feedbacks uh give us like some past data and it feeds it back to the future or the present so we have a ubiquitous feedback here we have some

feed forward elements we have elements of feedback that do certain things to the sequence and the question we ask here is what is the maximal number of steps needed to transform the sequence from 0 0 0 to one11 so we start with z00 Z we put that in we kind of move to our feedback loop our feedback loop does certain things to the sequence and then we end up with 111 and we actually want to know what the maximal number of steps is because we want to know you know how far we can go through these different uh feedback loops and get our answer this can also be restated in terms of temporal phases past present and future so we also have more explicitly this input of the present we have the future as an output and then we have these different uh Loops that give us the P give us past information and again we're interested in getting the most diverse information possible and so this is again this maximal Loop that we're looking for so we're trying to get the most variety out of these uh convolutional cybernetic architectures instead of trying to find the most efficient path or the tightest path or the most go directed path we can further understand this or restate this in terms of semantic components so we can talk about this in terms of symbols reference and understanding symbol being the input reference being the feedback or part of the feedback loop and then understanding being the output and so we want to accumulate as many reference as

possible for a symbol so that we get an output of understanding and again it's the variety that counts not optimizing the process or trying to find a goal directed aspect to this so if we think about how these ccas are assembled uh we can look at the different components of feedback and then we can start to assemble these more complex structures so let's look at first order feedback so CCA has input and in this case we're looking at past present future and maybe alternative Futures and things like that and they all produce a predictive output so first order feedback would be an input going to a future going to the present and naturally the future is just like the present plus the past the present comes in as in it's passed to this feedback which is the past the present and past are added to the Future and then the output is some sort of predictive input so again we have this reordering of past present future in the first order feedback if we have a feed forward only case we have input which leads the present which leads to some complex output but of course as a feed forward only system we just only get the benefit of having the present we don't get to talk about the past in the future it's just like every input results in a an output and that output then is complex output within any context then if we put together a larger set of feedback loops we can see this sort of convolved mixed

feedback where we have first order
feedback second order feedback and a
feed forward component so these are all
different components producing two
different complex outputs so we have an
input that gives us the present we get a
first order feedback
that gives us some past and we add those
together to a future state that produces
complex output we can also have a feed
forward part of that where the input
goes to the present and then the present
isn't part of that state and then we
might have a a a second order feedback
where we have an input which is the
present it passes A first order feed
Loop uh feedback loop to the past and
another first order or second order
feedback grp to the past and those can
both be combined into the present which
is then informing a future state which
involves a longer memory which has more
information and that gives us another
complex output but we also have this
path from the second order feedback up
to an alternative future which then gets
passed out as a complex output so you
see what we have here is we have an
input it goes to the present we Sub sub
reference the past we sub reference the
past again and that gives us an
alternative future so this happens for
every cycle so you end up with these
three different types of complex
output and given this topology we can
end up with these different types of
characterizations of the system remember
again with this convolution what we're

introducing is this variety which is actually a concept from uh the ever good regulator theorem and a lot of the work of W. Ashby so we're dealing with producing the maximum amount of variety so we you know we need variety to control systems so this is actually overall the appearance of purposes replaced with an ensemble of positive and negative feedback elements with diffuse effects so remember that we need to have not only a single closed loop feedback but we need to have multiple feedbacks and maybe some open loop feedbacks in there so we can have uh maximize a variety so the imperatives of such a network are to gain processes or to maximize the number of processes and this leads us to maximizing variety and of course variety was a concept that was introduced by W. R. Ashby in the ever good regulator theorem and he talked about how variety is necessary for control but it also creates the conditions for adaptability modularity and modifiability so we have systems that are both not purposeful but they also have this sort of imperative for control but also for other types of conditions in a system that allow it to adapt be modular and be modifiable and so we can also add in polarity we can have different types of effects resulting from feedback as we talked about if we reorder past present and future that's all well and good but this also has relevance to different

types of processes that result in systems so in in complex systems we have these different effects like Avalanches runaway feedbacks and dampened feedback these are non-linearities and we saw an example of avalanches earlier in our experimental cybernetics and and that can result from a number of feedbacks of different polarity and this can produce unpredictable inst stability we can have things like runaway feedbacks which are basically just the product of positive feedback and that can lead to mass and stability and we can also have something like dampened feedback which is a negative overly negative feedback which sometimes makes things more controllable and so we can have these effects that result from these processes that have a polarity that lead to these different complex systems phenomena that then lead to things that are maybe look they look te logical or purposeful but they're actually just the products of complexity and so if we kind of go back and look at like how we can model things like connectionist learning and reinforcement learning with cybernetics we can actually take the equations of connectionism and reinforcement learning and put them into this sort of cybernetic system system so this is where you know we want to take modern learning systems or artificial learning systems and understand them using cybernetic approaches and maybe apply them into these kind of

ccas and so this this is congruent with
hopefield and Sutton's alternative
Viewpoint which is that the weights of a
network are reinforced by the activity
and selection of previous inputs
so this is where we have we're looking
at Q learning here we're looking at
connectionist learning and we're looking
at sort of the ways equations are
structured and we're thinking about how
you know we can get learning from these
systems of equations and you know this
is a dynamical system you know as much
as a learning system so we want to
understand kind of how this is how this
works and so we started a random set of
weights and lack structure due to
selection we started a random set of
weights and lack of structure due to
selection we proceed to a structured set
of weights and optimal policies
unwittingly we kind of move from a
markovian system to a non-markovian
deterministic
system and so there's a tri question
here which is how do we go from
something that does not resemble purpose
to something that is purposeful the
answer is is that purposes a post talk
determination so we can model this as
habits with weights so we have this
input we have these outputs and we have
this uh CCA that's that has polarity and
we can have habits that are the serve as
the inputs and the weights are the
polarity and we have non-ic behavior in
a directed graph so this directed graph
kind of moves towards the end points the

outputs and so the lesson here is that purpose is basically whatever we determine these outputs to be so when we observe a behavior a behavior is generated by the CCA

graphs we interpret the output the way we'd like to interpret it and as I said before we can have these different outputs that are that look purposeful or they look peely logical but that's just really a post talk

determination and we move to Wier's Alternative Viewpoint so Wier is more of a cybernetics guy he also has an alternative

Viewpoint and his Viewpoint focuses on communication and information which have both a control layer and a semantic layer so again we have to go back to the past present and future formulation so instead of thinking about this in terms of polarity we want to think about this in terms of not just the control but the semantics as well and so we have this past present and future we have an input which is the past

we have a future which is the present with past feedback we have the present so the present goes to the Past goes to the future but the future also moves to a present where there's a present with past and future components and then an output so if we run this in time if we run this as a dynamical system we end up with this the present being sort of you know it's not really additive but it's basically concatenating things onto the present with information from the past

and you're informing a future trajectory
and you're you can actually generate
alternate futures of a system from this
and so this there's a control layer a
semantic layer and this gives us
information about the system both in
terms of contingencies and synergies so
connectivity and ordering provides both
contingencies
which is that nodes depend on other
nodes and synergies for nodes working
together so in this model we can't knock
out any one node and have the system
work we can't knock out the the feedback
and have this sort of
Rich self- reference we can only have
you know a feedforward model and we've
seen that that's much less
interesting we also can't knock out the
future and we can't knock out the
present we have to connect continue with
all of our
nodes but also these nodes working
together produce the synergistic
um trajectory that gives us rich
information about time it gives us this
present it gives us a a future but it
also gives these intermediate states
where we get past feedback and we get
integration of
those this stands in contrast to the
imperative of purpose and
heliology is cybernetic control and beh
or purposeful or stochastic activity
tempered by recursively recombined
temporal ordering and so this is again
one of the
ccas and a CCA is also a series of steps

that add recursive functionality so sometimes if we think about this in terms of our internal model memory this is memory with sub referential power so what allows us to have a memory again where we can reference the past of a past of a past and it informs the future of the present and adding present into future and things like that this creates a hierarchical system and high degrees of intentional order so again we can add these things together or we can run these ccas and we can produce this memory with sub referential power and the output can look very organized and hierarchical but the thing that produced that output is not necessarily ordered and it's not necessarily uh there's no overarching goal or any other sort of information going into it so once again thinking about habit and purposeful Behavior we have this paper creatures of habit the Neuroscience of habit and purposeful Behavior this was published in biological Psychiatry so this is where they talk about animal behaviors being composed of Habitual and gold directed categories habitual is non-teleological go gold directed is theological they provide this sort of breakdown where you have theological versus nonta logical behaviors and deterministic versus stochastic Behavior if we think about ta logical and deterministic behaviors we get gold directed behaviors those are things like reaching for a Target or

moving towards a goal but of course we know that some of these types of goal-seeking behaviors are chemo like chemotaxis or phototaxis are stochastic and so if we have teleological and stochasticity we might have something like chemotaxis or phototaxis we have something it's non-chological and deterministic we have habitual work so we have something that's habitual it's not moving towards a goal but it's deterministic something like uh scratching or like a reflex you have nonic on stochastic behavior however we know that that looks something like a random walk so we can actually characterize these different behaviors in this way the problem with chemotaxis and I put a question Mark in the table so chemotaxis for example which is a movement along chemical gradient we already established that that was not t logical the goal directedness in that case is due to the environment not the internal model and so we can look to Gibsonian information or we can look towards sort of thinking about why it's not te ological and so but it doesn't also doesn't resemble a random walk so that provides us with a problem which is where does this fit into this typology so a clue to kind of where chemotaxis and phototaxis fits into that typology comes from this paper brain like neurodynamics for Behavioral control develop the reinforcement learning and this is from the bioarchive

from this year total at all test two
scenarios supervised or extent and
reinforcement or developmental learning
so we have extent and developmental
learning only reinforcement learning
produces goal directed behavior and
neuroactivity similar to biology it
provides robust short-term behavioral
adaptations to
perturbations and it produces an
emergent emergence of policies that are
not purposeful in and of themselves but
lead to things that look the part so
this is reminiscent of our CCAs it's
also reminiscent of our to our
discussion of chemotaxis and phototaxis
building a CCA from a small Motif is a
form of purposelessness or
purposelessness reinforcement learning
build up you build up to function and
sometimes useless structures and that's
okay another example of nonpurposeful
behavior that kind of fits into our CCA
model comes from feudal substrate Cycles
in Biochemistry and so this is the paper
stochastic amplification and
signaling enzymatic feudal Cycles their
noise induced by stability with
oscillations this is from
PNAS so this is where we describe these
feudal Cycles is these feedback loops
where you have an input of energy
there's entic
reaction and you get nothing that comes
out except entropy or heat so you
basically don't have an output your
output is entropy it's heat and of
course this is a problem because we

expect there to be an output we expect
this reaction to actually do something
and in fact it's creating heat but it
seems like that's a lot of work for just
heat so we have a biochemical pathway
which is a close feedback it does
thermodynamic work for no
reason heat is produced from an
energetic input but we don't get any
biochemical
products and this seems like it's a
waste of energy that has no
purpose but it's actually important for
regulation and other processes it
produces complex higherorder Dynamics or
amplification effects throughout a
greater biochemical system so if we go
from convolution to order we can see
that the evolution of cybernetic control
is a causal recursive chain of
contingencies and so here we look at the
contingencies in a CCA we go from one to
two to three and then that's a feed
forward system and then we can go from 1
to two to three prime to four which
feeds back to two and gives us this
output four prime or we can go from 1 to
two to 3 to four or 1 to two to three
prime to four to 5 to 6 to 7 which is an
output so we have these different
outputs from these different numbered
boxes and we have these uh resulting
processes so let's take three prime as
an example three prime is dependent on
two and one and once feedback is
operational four and five so four and
five actually do feedback to three prime
which of course is a dependency so if

you knock that dependency out you lose
your entire functionality of that part
of the
network seven on the other hand which is
another output at the top is dependent
upon 6 54 3 Prime 2 and one there's no
feedback from
delay so you can see the different types
of order and how there's a lot of
recursive chains that lead to
contingencies so again if we think about
our feudal cycle our feudal cycle is
like something at the bottom I her or
feedback back is like something at the
top of this diagram and our alternate
outputs alternate outputs are at the
right and they are of course maximizing
for variety and so if we think about the
evolution of cybernetic control as a
causal recursive chain of contingencies
we can think about the kinematic chain
which is where we have these
dependencies so D is affected by the
activities of ab and C we get this chain
it's this is this robotic arm that
extends out across two joints and the
activity in D with kinematic is
constrained by A and C and so we can
describe it in that way if we go back to
these kinds of
contingencies we know that seven is
constrained by its intermediates and
thus it is with the kinematic
chain in in the kinematic chain we can
do this in a linear fashion we can add
on conditionals as convoluted cyber as
the convoluted cybernetic Network
evolves this restricts the system to a

deterministic path behaviorally this leads to interesting consequences for developing sensory motor Loop and a kinematic chain is basically sensory motor physics with contingency so here are examples of a kinematic chain in robotics and we can think about uh embodied robotics here where we have these control systems for these kinematic chains maybe the CCA where we have different types of morphologies and we can design them appropriately we can design and we can design them appropriately we can design behaviors that aren't purposeful aren't teleological but still give us behaviors that we can that look intelligent we can also think about purposeful behavior and fore cognition so our approach to purposeful Behavior can be applied to Fore cognition so we have these different types of cognition some of them being embodied some of them being embedded some of them being extended which means outside the brain and some of them being inactive if we put this in the framework of Rosenbluth all's behavioral classification does for Fore Force us to rethink that behavioral classification as we did in a number of instances in this talk now there's one Insight of 4E that's interesting to us which is that the morphology environment of a system is more important than its internal model so we've been talking about internal models

and how these ccas model the internal state of behavior and how that shows that in a lot of cases Theology and purpose are sort of elery but if morphology and environment are more important than the internal model then maybe we need to incorporate our ccas uh or or recast our ccas in terms of morphology and environment as well as an internal state in fact our CCA do that to some extent but it's uh sort of a a way to put this behavioral classification on its head and so in the case of 4E purpose is not ascribed to a mind feedback is diffuse and complex prediction its highest orders are not theological they must be open-ended we talk about this in a paper from 2023 and Royal Society interface Focus so what is behavior and what are its properties so yeah we we're kind of revisiting what Rosen Bluth talked about we're looking at a number of examples from neuroscience and from psychology and from other areas animal behavior Robotics and we kind of break down kind of what we're looking at here so Behavior can be intentionality it can be goal directed and supervised and so these are things like control systems par cognition and of course some of those are go directed some of those are regulated complexity then we have Rich higher order complex Dynamics and those are things like Bey reactions which we talked about an embryogenesis which we

didn't really talk about but also qualifies in that category so these are D outputs of dynamical systems we also have connectivity and convolution which are things like connectionism avalanches and open-ended learning and then finally we have behaviors that are hardwired but recursive so with our ccas we were modeling some Hardware behaviors in a recursive fashion we also talked about this in terms of non-markovian systems and habitual Behavior we want to go beyond the standard of the logical and purposeful Behavior so a standard way of thinking about behavioral regulation is that inputs are deterministic and outputs or Collective output should be close-ended so if we expand the basic Loop feedback model we get an interconnected network of retrocausal ordering so this is something that we're seeing in the CCA where we talk about the order of causality past present future we have this basic loop feedback model but if we expand that out we get much richer retrocausal ordering which leads to of course to variety but it also leads to these interesting paradoxes and interesting effects of higher order feedback behavior is often open-ended and this can be characterized by embodied Dynamics unpredictable which can be characterized by a chaotic Dynamics nonnormal which can be characterized by non-markovian behaviors and systems

Avalanches of different types power law
Dynamics and so forth and
variational which means that it's
characterized by multiple goals and
redundancies and so this requires a
reassessment Rosen blue teal
typology and has wider consequences for
the practice of active
inference thank you for your
attention uh one of our co-authors Jesse
parent uh produced a lengthy blog post
on the Rosen blue that all paper talks
about its enduring insights and its con
uh its context in
cybernetics and then we have two papers
we talked about in the talk from our
group that stasis
machines and a primer on Gibson need
information so I thank you for your
attention
e for